

Analyzing Data

- 1 Motivation
 - temperatures in Chicago
- 2 Gathering, Storing, and Plotting Data
 - the national weather service
 - from text to array
 - making plots
- 3 Data Frames
 - spreadsheets for programmers
 - sorting data
- 4 Data Projects
 - more proposals of topics for the first project

MCS 472 Lecture 5
Industrial Math & Computation
Jan Verschelde, 23 January 2026

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temperatures in Chicago

Are the winters in Chicago becoming milder?

We reformulate this question with three other questions:

- 1 Is there data to address this question?
- 2 How to acquire, store, and represent the data?
- 3 Does the original question become a mathematical one?

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the national weather service

The National Oceanic and Atmospheric Administration (NOAA) has a site `https://www.weather.gov` (The National Weather Service):

- 1 Navigate to Past Weather.
- 2 Enter Chicago, IL as the location.
Click Chicago on the map at
`www.weather.gov/wrh/climate`
- 3 Make the selections to request data.

See the screen shot on the next slide.

selecting the mean temperatures

<https://www.weather.gov/wrh/climate?wfo=lot>

NOWData - NOAA Online Weather Data

1. Location »
[View map](#)
Chicago Area
Rockford Area
Antioch, IL
Barrington 3 Sw, IL
Chicago Botanica, IL
Chicago Midway 3, IL
Chicago Northerly, IL
Chicago O Hare AP, IL
De Kalb, IL
Dwight, IL

2. Product »
☐ Daily data for a month
☐ Daily almanac
☒ Monthly summarized data
☐ Calendar day summaries
☐ Daily/monthly normals
☐ Climatology for a day
☐ First/last dates
☐ Temperature graphs
☐ Accumulation graphs

3. Options »
Year range: 1922 - 2021
Variable
Avg temp
Summary
Mean

4. View »
[Go](#)

Looking at the monthly mean temperatures for the past 100 years...

retrieving data in text format

No plain text file is directly available to download ...

There is a hack:

- 1 Copy and paste the numerical data.
- 2 Make a header line separately and pasted in.
- 3 The file `tempChicago100years.txt` stores the data.

Copy and paste is not ideal, but still better than manual data entry.

a first exercise

The lecture continues with the average temperatures.

Another question related to the original question is

Are the winters in Chicago becoming more extreme?

To address this question, solve the next exercise.

Exercise 1:

Download the minimum and maximum monthly recorded temperatures in Chicago for the past one hundred years.

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making an array in a Julia session

```
julia> using DelimitedFiles
```

```
julia> A = readdlm("tempChicago100years.txt")
```

```
101×14 Matrix{Any}:
```

"Year"	"Jan"	"Feb"	"Mar"	..	"Dec"
1922	24.8	29.4	39.3		29.9
1923	30.7	22.3	33.0		39.7
.					
.					
2019	21.0	25.9	34.3		34.0
2020	30.1	30.2	42.8		32.8
2021	29.2	20.2	44.2	..	38.0

```
julia>
```

The output of the session is edited for brevity.

- The numbers on file were separated by spaces (not commas).
- The type `Any` was caused by the header (strings).

the session continues ...

```
julia> using Statistics
```

```
julia> A[1,13]  
"Dec"
```

```
julia> dec = A[2:end,13];
```

```
julia> mean(dec[1:50])  
29.223999999999997
```

```
julia> mean(dec[51:end])  
28.671999999999997
```

```
julia> mean(dec[1:10])  
30.04
```

```
julia> mean(dec[91:end])  
32.06
```

two more exercises

Exercise 2:

Compare the mean temperatures of the ten decades in the past 100 years for the four winter months.

Exercise 3:

For the minimum and maximum temperatures downloaded in Exercise 1:

- 1 Compare the means of the minimum and maximum of the ten decades in the past 100 years for the four winter months.
- 2 Also compare the corresponding standard deviations.

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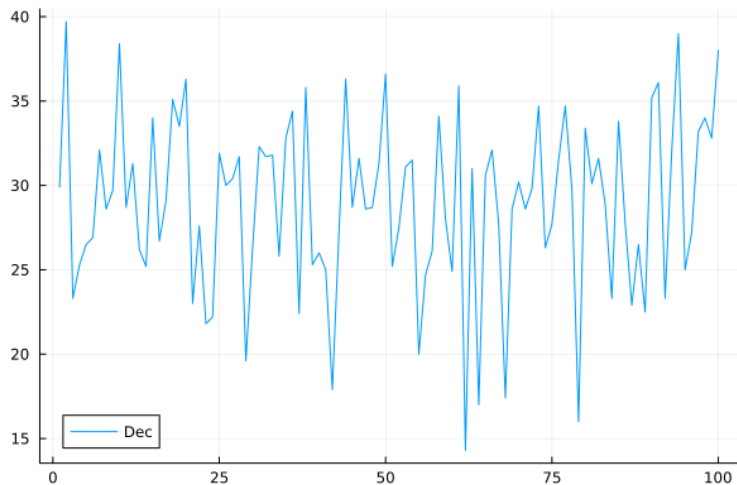
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the plot of the last 100 December months

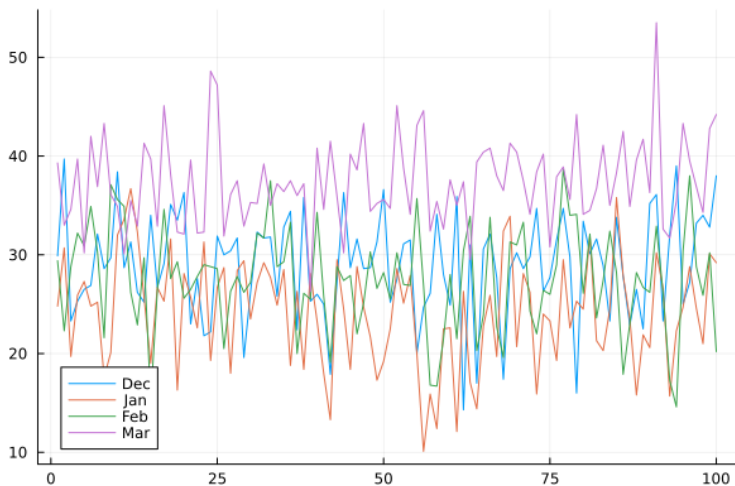


the Julia session continues ...

```
julia> dec[60:69]  
10-element Vector{Any}:  
 24.9  
 35.9  
 14.3  
 31.0  
 17.0  
 30.6  
 32.1  
 27.7  
 17.4  
 28.6
```

The eighties had some very cold December months.

the plot of the last 100 winter months



plots of monthly minima and maxima

Exercise 4:

For the minimum and maximum temperatures downloaded in Exercise 1:

- 1 Make a plot for the minimum and maximum temperatures for the four winter months for the past 100 years.
- 2 Make a plot for the *difference* between the maximum and the minimum temperatures for the same time ranges.

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data frames

Data structures are organized by dimension:

- 1 A *Series* is a one dimensional labeled array, capable of storing data of any type. The axis labels are called the index.
- 2 A *DataFrame* is a table with rows and columns.
 - ▶ columns may be of different type,
 - ▶ the size is mutable,
 - ▶ axes are labeled,
 - ▶ arithmetic can be performed on the data.
- 3 A *Panel* is a 3d container of data.

The name pandas is derived from Panel Data, as pan(el)-da(ta)-s.

The package `DataFrames.jl` is the Julia analogue to Pandas.

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our first DataFrame

```
julia> using DelimitedFiles
```

```
julia> A = readdlm("tempChicago100years.txt");
```

```
julia> using DataFrames
```

```
julia> df = DataFrame()
```

```
julia> df.Year = A[2:end,1]
```

```
julia> df.Dec = A[2:end,13]
```

This makes a DataFrame with the December temperatures.

the coolest and warmest Decembers

```
julia> sort(df, :Dec)
```

```
100x2 DataFrame
```

Row	Year	Dec
	Any	Any
1	1983	14.3
2	2000	16.0
3	1985	17.0
4	1989	17.4
5	1963	17.9
...	...	...
96	1971	36.6
97	2021	38.0
98	1931	38.4
99	2015	39.0
100	1923	39.7

The output is shortened to fit the slide.

one last exercise

Exercise 5:

Did the months with the minimum and maximum temperatures occur in the months with the smallest and largest means?

- 1 For the data downloaded in Exercise 1, make a DataFrame with the minimum and maximum temperatures for each December in the past 100 years.
- 2 Sort the columns and compare with the means.

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10. The Winters in Chicago

January 2023 appeared to be one of the warmest winter months, at least in Chicago, IL.

- Download data for the first winter month, starting December 21 to January 20, for the past 100 years.
Look for minima, maxima, and average temperatures for each day.
- Analyze the data with plots and segmentation, that is: divide the 100 years up in decades.

11. Earthquake Data

Which regions in the United States are most vulnerable to earthquakes?

- Visit `https://earthquake.usgs.gov` and download data.

Focus on three attributes

- 1 location,
- 2 time range, and
- 3 magnitude.

Make sure that for each location, the same time range and magnitudes are selected.

- Analyze the data with plots and statistics.
The software `Cartopy` (in Python, or through Julia) may be useful, but not necessary.

12. Displaying Geographical Data

Many data sets are geographical, e.g.: weather, earth quakes.

In this computational project, explore the software `Cartopy`.

- Summarize the features of `Cartopy`.
- Demonstrate the use of `Cartopy` on actual data.

bibliography

We computed the mean on downloaded temperatures, we plotted and sorted in a first data analysis.

Recommended sources:

- Wes McKinney: *Python for Data Analysis*, O'Reilly 2013.
- Jose Storopoli, Rik Huijzer, Lazaro Alonso: *Julia Data Science*.
First edition published 2021. <https://juliadatascience.io>
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International